

Rhys Lester-Talley

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EDUCATION

Columbia University, School of Engineering and Applied Science New York, NY
Candidate for Master of Science in Engineering: Computer Science, Software Systems, May 2027

Columbia University, School of Engineering and Applied Science New York, NY
Bachelor of Science in Engineering, Magna Cum Laude: Computer Science, May 2026
Cumulative GPA: 4.08/4.00 | Tau Beta Pi Inductee, Upsilon Pi Epsilon Inductee

WORK EXPERIENCE (SELECT)

Software Developer Intern, IBM Corporation (Poughkeepsie, NY, 05/2025-Present)
Worked with IBM z/TPF's middleware and security team as well as their database squad. Implemented middleware monitoring for TLS certificates, resolved critical errors which caused shared system sockets to hang, and interfaced directly with clients to assess requirements, redesigning projects to meet their needs. Interned from Summer 2025 until present, converting a full time offer for June 2026 into an additional internship in order to pursue a masters.

Software Engineering Intern, Columbia Build Labs (New York, NY, 09/2024-05/2025)
Matched with [AMLAH](#), an early-stage tech startup revolutionizing real estate markets with remotely guided tours. Focused on backend development and system design, developing data models, database schemas, and RESTful APIs for use in managing and optimizing the scheduling of remote tours, agent availability, and inventory management.

TA for Computer Science, Columbia University (New York, NY, 01/2024-Present)
Serve as a TA across four different classes for three different professors: Introduction to Java, Data Structures, Advanced C Programming, and Design with C++. I work with Pygrader to develop homework rubrics and grading frameworks, answer questions in office hours and on online discussion forums about class concepts and coding best practices, and design exam questions, homework assignments, and projects about course material.

Research Intern, DitecT Lab at Columbia University (New York, NY, 02/2024-09/2024)
Worked with Professor Xuan Di at Columbia's DitecT Lab on the WayCARE project. Expanded on existing codebase to deploy and update full stack weather data dashboard.

SKILLS AND PROJECTS

Technical Skills: C, C++, Java, Assembly (x86, MIPS, Assembler), Python, SQL, Git, VSCode, Codex/Claude Code.

Remote TLS Certificate Expiration Monitor

Implemented new compliance tooling in IBM's proprietary z/TPF operating system to store and monitor remote TLS certificates using C, C++, and Assembly for the benefit of large clients such as Visa, Amtrak, and the IRS. After TLS handshake, remote client/server certificates are stored in a hashmap within system heap memory. A custom command displays all certificates currently stored in the table as well as their expiration date. Furthermore, once per day all certificates expiring within a configurable amount of time are processed via a customizable user exit.

Parallel I/O and Compression Migration

Designed and developed a migration routine to convert a hierarchical, record oriented TPFDF database into an indexed, compressed structure enhancing read times by up to 96%. Leveraged existing CRUISE architecture and DF APIs to keep code repetition to a minimum and added enhancement to allow for fallback migration.

INTERESTS

Interests: OS and systems programming, video games, weightlifting, soccer, fantasy novels.